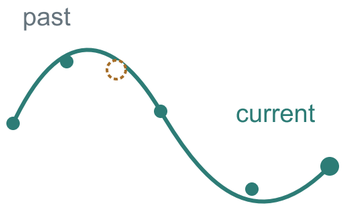
Inside the Track MLP

An ordered 16-frame 3D history becomes one 512-D token per point

1 One point history

Point i across time



$x_i(0)$: current XYZ anchor

XYZ: 16 × 3

Validity: 16 Time: 16

History: −0.5 s to 0

2 Eight features per time slot

Anchor 3 channels	Motion 3 channels	Time 1	Valid 1
x_0 / 0.5	Δx(τ) / 0.05	−0.50	1
x_0 / 0.5	Δx(τ) / 0.05	−0.47	1
0 0 0	0 0 0	0	0
⋮	⋮	⋮	⋮
x_0 / 0.5	Δx(τ) / 0.05	−0.03	1
x_0 / 0.5	0 0 0	0	1

16 × 8 matrix · missing slots are all zero

Flatten
16 × 8
→ 128

3 Flatten, then apply the shared MLP

Linear 128 → 128
16,512 parameters

ReLU

Linear 128 → 128
16,512 parameters

ReLU

Linear 128 → 512
66,048 parameters

Point token
$p_i \in R^{512}$

99,072 trainable parameters

Next: LayerNorm
→ fusion keys / values

Exact feature construction

Effective validity: $\tilde{m}_i(\tau) = m_i(\tau) \wedge m_i(0)$

$f_i(\tau) = where(\tilde{m}_i(\tau), [x_i(0)/0.5, \Delta x_i(\tau)/0.05, \tau, 1], 0)$

$\Delta x_i(\tau) = x_i(\tau) - x_i(0)$

Positions in meters; time in seconds. Validity distinguishes missing data from zero motion.

Shared across points; time is flattened first

K × 128 → **Same MLP weights** → **K × 512 tokens**

No cross-point mixing or point-ID embedding inside the MLP.

Current-invalid tracks are excluded before fusion attention.

Stationary control: zero only past displacement; preserve current anchors, timestamps and validity.

Current setting: H = 16, K = 16, token width = 512 · input trajectory is schematic